

Question Paper

Exam Date & Time: 20-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: MAT 2122/MAT_2122 ENGINEERING MATHEMATICS - III

(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Find the Fourier series expansion of (3)

$$f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi < x < 0 \\ x - \frac{\pi}{2}, & 0 < x < \pi \end{cases}$$

with $f(x + 2\pi) = f(x)$.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Express $f(x) = 2 + x$ as a half-range sine series in $(0,2)$. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Obtain the Fourier series expansion of up to the second harmonics from the following table. (3)

x	0°	60°	120°	180°	240°	300°	360°
y	3	5	5.7	6.3	7.2	4.3	3

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Find the Fourier cosine transform of the function (3)

$$f(x) = \begin{cases} 4x, & 0 < x < 1 \\ 4 - x, & 1 < x < 4. \\ 0, & x > 4 \end{cases}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2B) Find the homogeneous system whose solution space W is spanned by the set of vectors (3)
 $\{(1,0,2,3), (2,3,1,4), (3,3,5,7)\}$.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) $\begin{bmatrix} 0 & -1 & 1 \end{bmatrix}$ (4)

Diagonalize the matrix $A = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$.

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) Solve the following system of equations by LU decomposition method. (3)

$$\begin{aligned} x + y + z &= 1 \\ 4x + 3y - z &= 6 \\ 3x + 5y + 3z &= 4 \end{aligned}$$

Course Outcome: "CO3" / Cognitive Level: "Application"

3B) Find the Cholesky decomposition of (3)

$$A = \begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix}$$

Course Outcome: "CO2" / Cognitive Level: "Application"

3C) Find a singular value decomposition of $A = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix}$. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

4A) Let W be the subspace of $\mathbb{R}^4$ spanned by the set of vectors $\{(1,2,0,1), (2,1,0,3), (3,1,2,4)\}$. (3)

- (i) Find a basis and the dimension of W .
- (ii) Extend the basis of W to a basis of $\mathbb{R}^4$.

Course Outcome: "CO3" / Cognitive Level: "Application"

4B) Let W_1, W_2 be the subspaces of $\mathbb{R}^4$ given as (3)

$$W_1 = \{(a, b, c, d) / b - 2c + d = 0\} \text{ and}$$

$$W_2 = \{(a, b, c, d) / a = d, b = 2c\}.$$

Find a basis and the dimension of

$$(i) W_1 \cap W_2 \quad (ii) W_1 + W_2.$$

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4C) (4)

Find a basis and the dimension of kernel and image of the linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T(v) = Av$,

$$\text{where } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \text{ and } v \text{ is a column vector of}$$

$$\mathbb{R}^3.$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Find the matrix representation of the following linear operator T on $\mathbb{R}^2$, defined as $T(x,y) = (2x - 7y, 4x + 3y)$, relative to the basis $S = \{(1,3), (2,5)\}$. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Let $V = C[a,b]$ be the vector space of continuous functions with the inner product $\langle f(t), g(t) \rangle = \int_a^b f(t)g(t)dt$. Determine the cosine of the angle between the functions $f(t) = \cos t$ and $g(t) = \cos 2t$ in $[-\pi, \pi]$. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Let W be the subspace of $\mathbb{R}^5$ spanned by $u_1 = (2,2,-1,0,1)$ and $u_2 = (-1,-1,2,-3,1)$. Find a basis of the orthogonal complement $W^\perp$ of W . (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 MAT_2122:ENGINEERING MATHEMATICS-III

#	Course Outcome	Description	Marks
1	1	CO 1	13
2	2	CO 2	11
3	3	CO 3	12
4	4	CO 4	7
5	5	CO 5	7
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 MAT_2122:ENGINEERING MATHEMATICS-III

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	13	43(86%)
4	Analysis	2	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)